Integrating a plugin to Hub Platform

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Basic information

Environments

There are 4 environments: **dev**, **qa**, **stg** and **live**. Hub Platform development process is based on:

1. Having all latest features (latest firmware and plugins) in **dev** environment
2. When features are ready for testing they are moved to **qa** environment
3. When it's time to release new features they are moved to **stg** environment (where pre-release testing is preformed)
4. If everything is good, features are moved to **live** environment (it is used to deliver new firmware and plugins to end user controllers)

Hub types

Hub Platform (Linux Firmware) is used on multiple controller types:

1. Vera Edge (g150)
2. Ezlo Plus (h2.1) *(may be also referenced as e450 in some docs)*
3. Ezlo Secure (h2_secure.1) *(may be also referenced as e550 in some docs)*
4. ...

Atom based controllers support same API for plugins, but have less resources (cpu/memory). Lightweight plugins can be used on both Atom and Hub Platform based controllers. But plugins for Atom based controllers are distributed differently, and this topic is out of scope of this article.

Each controller type have differences in hardware and capabilities, so it is possible to make plugin available only for limited set of controller types.

It is also possible to specify if plugin will be installed on specific controller type by default.

OpenWRT

Hub Platform (Linux Firmware) is based on OpenWRT. Both firmware and plugins are distributed as OpenWRT packages.

Packages are distributed through OpenWRT packages repository. Roughly each environment can be considered as a separate OpenWRT repository.

So in order to integrate new plugin to Hub Platform you have to:

1. Build OpenWRT package for your plugin
2. Deploy plugin package to Ezlo public server.
3. For each environment and hub type:
   a. Specify if plugin should be available for installation.
   b. If plugin should be available, specify what version of plugin should be used
   c. Specify if plugin should be installed by default
4. Trigger rebuilding of corresponding OpenWRT repositories.

Other information

- Plugins (plugins versioning, dependencies, etc...)
- Plugins Architecture (more topics on plugins)
- Software WS API (API to install/update/uninstall plugins)

Integration Manual

Building and deploying OpenWRT package

We've created a git repository with utils that should be used to build and deploy plugin packages: https://github.com/ezloteam/HubPluginsUtils

If you are using git to develop your plugin, you may include it as git sumbodule.

As for now this repository contains python3 scripts for building and deploying a plugin. You can find details about these scripts at: https://github.com/ezloteam/HubPluginsUtils/tree/master/opkg

Later on more helpful utilities will be added to that repository.

Feel free to create Pull Requests with existing utils improvements, or new utils that can be useful for other plugin developers.

Configuring OpenWRT packages repositories

When new plugin package is deployed to server, it is not yet available for installation or upgrade on controllers. You have to include it into OpenWRT repository.

We've created a git repository that contains configuration of OpenWRT repositories for each environment and hub type: https://github.com/ezloteam/HubPlatformBundle

This repository have Jenkins Job attached that will update OpenWRT repositories based on config.json file.

So you have to create Pull Request to a branch corresponding to environment that you want to update with corresponding changes in config.json file.

When Pull request will be approved it may be merged. After merge Jenkins will rebuild corresponding OpenWRT repository (~10 minutes), and your plugin will be available for installation or upgrade for controllers with specified environment firmware.